COMP 441 — Software Analysis and Testing

Week 10 — Metrics, Defect Density and RTM Completeness

Group A · Track D · TOML Kit

Period	Week 10
Evidence folder	reports/week10
Repository	github.com/the-dezeray/tomlkit
Commit(s)	6b8884f
Generated	2026-10-27
Contents	5 sections

This pack collects the week's human-readable report and key evidence. Raw AI-only logs are listed in the inventory and intentionally not printed.

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Source: reports\week10\metrics_log.md

Week 10 Metrics Log: tomlkit (Track D)

Group project: COMP 441, AI-Augmented Test Suite Repository: the-dezeray/tomlkit (fork of python-poetry/tomlkit)
Environment: Python 3.12.10, pytest 7.4.4, mutmut 2.5.1, hypothesis 6.129.5, radon 6.0.1

Step 2: Test suite run

- Date: 26 October 2026
- Command: `poetry run pytest -q tests`
- Result: 1058 passed, 0 failed (baseline, before new tests)
- Duration: 6.86s
- After new tests: 1063 passed, 1 xfailed (tests/test_types.py, test_properties.py)

Step 3: Coverage

- Command: `poetry run pytest --cov=tomlkit --cov-branch --cov-report=term --cov-report=html tests`
- Tests: 1058 passed in 23.53s
- Statements: 3330 total, 353 missed, 2977 covered
- Statement coverage: 89.4% (2977 / 3330)
- Branches: 1523 total, 1282 covered, 241 missed
- Branch coverage: 84.2% (1282 / 1523)
- Combined coverage (coverage.py "Cover" column): 88% (not used in the report)
- Excluded lines: 12 (not counted in either figure)
- Weakest modules: _types.py (57%), items.py (81%), container.py (88%)
- Raw data: reports/week10/coverage.json, htmlcov/index.html

Step 4: Mutation testing

- Scope: tomlkit/_types.py only (the other modules have not been mutated yet)
- Runner: python -m pytest -x --assert=plain tests (tests/ only, 1058 tests)
- Mutation run 1 (before new tests): 1 killed, 9 survived, 10 total
- Mutation run 2 (after tests/test_types.py): 4 killed (mutants 6, 8, 9, 10), 6 survived (mutants 1-5 equivalent; mutant 7 is a test gap, defect M1)
- Mutation score (brief formula): 40% (4 / 10 x 100)
- Adjusted score (excluding 5 equivalent mutants 1-5): 80% (4 / 5 x 100)
- Week 7 (v1) mutation score for comparison: 62.0% (3087 killed / (3087 killed + 1890 survived) x 100; 26 timeout/suspicious excluded); modules: _utils, api, container, exceptions, items, parser, source, toml_file (no overlap with Week 10 scope, which is _types.py only); runner: [RUNNER]; not directly comparable to Week 10
- Limitation: mutation testing covers one module; the library-level score is not yet measured

Step 5: Defect density

- Lines of code: radon raw -s tomlkit, total SLOC = 4158 (LOC = 6197)
- KLOC: 4.158
  - D1: date() accepted datetime strings (fixed, commit a4f54db)
  - D2: Bool.**eq** returns NotImplemented for another Bool (open, xfail in tests_ai)
  - D3: inline-table duplicate raises KeyAlreadyPresent, not ParseError (open, xfail in tests_ai)
  - D5: key([]) serialises to an unparseable key (open, xfail in tests/test_types.py)
  - D6: comment() writes unparseable TOML for a bare CR (fixed, commit c829a38)
  - D7: bare CR in multiline strings (fixed, commit a38a3d8)
  - D8: insertion at position 0 drops the space after the inserted element (Week 9 session; open)
  - M1: _CustomDict.**or** drops existing keys because copy() returns {} (open, xfail in tests/test_types.py)
- Defect density: 1.92 defects per KLOC (8 / 4.158), provisional
- Note: D4 does not appear in any source available to the group; confirm with the Test Lead

Step 6: RTM completeness

- Total requirements: 16 (R01 to R16, week6/rtm/rtm_week6.csv)
- Requirements verified by a test of the requirement itself: 14
- RTM completeness: 88% (14 / 16 x 100)
- Excluded: R15 (listed tests do not exercise the D5 case); R16 (untested)
- AI suite result: poetry run pytest -q tests_ai -> 386 passed, 2 xfailed (D2, D3)
- Partial requirements (R03, R04, R12, R13): counted only where a listed test passes

Step 7: Property tests (Track D)

- File: tests/test_properties.py
- Hypothesis properties: parse reads back any TOML integer; dumps(parse(s)) == s for integer documents
- Result: 2 passed
- Covered functions: parse, dumps

Snapshot notice: Week 10 is an intermediate measurement. The final, authoritative numbers are in the final report §7 and `week11/mutation_v2/`. Where Week 10 differs (for example coverage 89.4/84.2 and v1 mutation 62.0% with timeouts excluded), the difference is scope or formula, not a correction.

Week 10 dashboard

Source: reports\week10\dashboard.csv

Indicator	Week 10 value	Notes
Statement coverage	89.4%	2977 of 3330 statements (coverage.py totals)
Branch coverage	84.2%	1282 of 1523 branches (coverage.py totals)
Mutation score (brief formula)	40%	4 killed / 10 total; tomlkit/_types.py only
Mutation score (adjusted)	80%	4 killed / 5 non-equivalent; tomlkit/_types.py only
Passing tests	1063	1 xfailed (known defect M1)
Total SLOC	4158	radon raw total
Defects found	8	D1 D2 D3 D5 D6 D7 D8 M1 (provisional; group log not yet consolidated)
Defect density (per KLOC)	1.92	8 / 4.158 (provisional)
RTM completeness	88%	14 of 16 requirements verified by a test of the requirement itself (R15 excluded; R16 untested)

Week 10 coverage summary (tests/ only)

Source: reports\week10\coverage.json

Metric	Value
Statements	2,977 / 3,330
Statement coverage	0.00%
Branches	1,282 / 1,523
Branch coverage	0.00%
Combined coverage	87.76%
Excluded lines	12

Consolidated defect log (final)

Source: reports\defects.csv

id	summary	status	severity	found_by	linked_test	commit
D1	date() accepted datetime strings	fixed	medium	AI baseline	tests_ai/test_ai_api.py::TestDateTimeFactories::test_date_rejects_datetime	a4f54db
D2	Bool.__eq__ returns NotImplemented for another Bool	open	low	AI baseline	tests_ai/test_ai_items.py::TestPickle	
D3	Inline-table duplicate raises KeyAlreadyPresent instead of ParseError	open	medium	AI baseline	tests_ai/test_ai_parser.py::TestInlineTables::test_duplicate_key_in_inline	
D5	key([]) serialises to an unparseable key	open	low	AI baseline	tests_ai/test_ai_api.py::TestKeyFactory	
D6	comment() writes unparseable TOML for a bare CR	fixed	medium	Manual mutation follow-up	tests_manual/test_kill_v1.py::test_k8_add_line_comment_rejects_any_line_break	c829a38

id	summary	status	severity	found_by	linked_test	commit
D7	Bare CR in multiline strings	fixed	medium	Property-based test	tests_property/test_properties.py::test_d7_multiline_string_with_bare_cr	a38a3d8
D8	Insertion at position 0 drops the space after the inserted element	open	medium	Exploratory session	tests_manual/test_kill_v1.py::test_k6_insert_at_index_zero	
M1	_CustomDict.__or__ drops existing keys because copy() returns {}	open	medium	Mutation testing	tests/test_types.py::test_custom_dict_or_keeps_existing_keys	

Raw evidence inventory

The pack prints the readable report and key summaries. Large raw artefacts are kept in the repository and listed here so a marker can verify them without paging through thousands of lines. Items tagged **[AI-only]** are deliberately not printed: they are the AI baseline's raw output, not the final hybrid result.

File	Size	Tag
reports\week10\coverage.json	41 KB	
reports\week10\dashboard.csv	673 B	
reports\week10\metrics_log.md	4 KB	